

Public Rubric

Monday, June 14, 2021 10:50 AM

How to use this rubric:

Go through each of the questions sequentially. Each question should be able to be answered with a yes or no. If any question is unclear or more background is needed, see the "Additional Details" and "Model Card Paper" sections. The first of these contains a description of what to look for in this category and the second contains a quoted description from the original model card paper (<https://arxiv.org/abs/1810.03993>) where possible.

All questions should be answered with respect to whether they are included in the model card. For example, if there is an ethical considerations section in a GitHub repository linked in the model card, answer no for Q20, but note that one existed in another source. If a model card links directly to a resource for a question, answer yes. For example, "to use our model, see the linked API (link here)." This would count as a yes for Q7. Do not read academic papers.

The questions should be answered as to simply whether or not the model card includes what is asked about in the question. For example, a model card narrowing the target distribution from the set of all strings to English sentences does describe the target distribution, so it should get a "yes." We would ideally want more information from the model card, but answering the questions this way is important to maintain consistency.

Category	Questions	Additional Details	Model Card Paper Quotes
Model Description	1. Does it provide a way to contact the model creator? 2. Does it list what type of model it is? • Ex: CNN, SVM, Bert, etc. 3. Does it list the model date/version? • Ex: Hugging Face versioning 4. Is a model license listed?	The model description should be a brief overview of model details. This is usually near the top of the model card. It should provide information about the model and its creators. This section is necessary to ensure that a model user is using the model they intend to use and has a means to contact the creator if an issue arises. The contact information must be explicit. For example, it can say "fill out this form with any questions." We will also count including an email as contact information. For example, "Author: Bob Smith, bsmith12@gmail.com."	Model Details. Basic information about the model. – Person or organization developing model – Model date – Model version – Model type – Information about training algorithms, parameters, fairness constraints or other applied approaches, and features – Paper or other resource for more information – Citation details – License – Where to send questions or comments about the model
Intended Uses	5. Does the model card explain scenarios in which to use the model? 6. Does the model card explain scenarios in which <i>not</i> to use the model?	The intended uses section of the model should inform the user exactly how to use the model. The user should gain a clear understanding of how they can and cannot use the model. This section is used to reduce improper usage of machine learning models. If not included, a model could do something unintended, potentially causing a variety of issues.	Intended Use. Use cases that were envisioned during development. – Primary intended uses – Primary intended users – Out-of-scope use cases
Usage Examples	7. Is a technical description of how to use the model provided? (interface, types, APIs, code examples, etc.)	Model cards are often provided when machine learning models are opened to public use. When this happens, it is necessary to explain how to use the model. This can come in the form of code examples or a link to separate API documentation. Some model cards exist for the purpose of transparency and are not intended for public use, so this section would be unnecessary.	
Target Distribution	8. Does the model card describe the target distribution (what inputs are in scope or out of scope?) 9. Are specific examples of any inputs or examples to illustrate the target distribution provided? • Ex: This model will not be able to recognize faces in blurry photos. An example of such a photo is provided below.	A model card may explicitly or implicitly mention their target domain. This is the subset of the domain that the model can be expected to perform well on. For example, if a facial recognition model is trained and tested on clear images then it cannot be expected to work on blurry photos. If the model card mentions that there will be performance issues on blurry photos, it is describing the target domain. Some model cards may discuss this in the section on training, saying "this model was trained on clear images." For consistency, we consider any model that narrows the inputs beyond just types as describing the target distribution. For example, saying a model is used to answer questions in French would be a yes. This is because the model card narrows the input from strings -> French strings -> French question strings. Ideally, we would want this model to give us examples of question types that it will accept or will not accept. Take note of the level that the target distribution is described. See this paper for more information: https://www.datascienceassn.org/sites/default/files/Shortcut%20Learning%20in%20Deep%20Neural%20Networks.pdf.	
Evaluation Metrics	10. Are evaluation statistics reported in the model card? 11. Are the meanings of these statistics explained and / or justified? 12. Are graphs / other visuals that demonstrate model performance presented?	Most machine learning models are not made to solve a new problem. There are likely dozens of other models that perform the same task. In order to persuade users to use their model, model card creators often list evaluation metrics. These can provide the user with a sense of how well this model will work on their real-world data. Any metric that is listed should be explained in context. For instance, saying a classification model has 98% recall does not mean anything to someone who has never heard of this statistic. On top of this, one would need to know how the model creators defined a true positive, false positive, etc. to understand the actual implication of this statistic. It can also be useful to have visuals that go along with these statistics.	Metrics. Metrics should be chosen to reflect potential real world impacts of the model. – Model performance measures – Decision thresholds – Variation approaches
Evaluation Process	13. Is the model evaluation process explained? 14. Is the evaluation data	To maintain transparency and establish credibility in the evaluation metrics provided, the evaluation process should be explained. A good explanation should include how the model creators gathered the evaluation data, any potential problems with the evaluation data, and	Evaluation Data. Details on the dataset(s) used for the quantitative analyses in the card.

	explained? 15. Is the evaluation data available? • Ex: The Top Word List Dataset (answer no if not linked or reproducible)	how they used this data to evaluate the model. This section encourages the reproduction of results. If it is possible to provide the raw evaluation data, it should also be provided.	– Datasets – Motivation – Preprocessing
Training Process	16. Is the training process explained? 17. Are properties of the training data explained? 18. Is the data collection or creation process described? 19. Is the training data available?	The training process section goes hand-in-hand with the evaluation process. This section should provide details about how the model was trained. Any potential issues with the training data should be explained in this section. Details about the training data should be provided such as how it was gathered and what kind of data it is. Properties of the training data can include any information that would be useful to a potential user of the model. For example, some properties could be how many data points were used for training, the type of data used for training (ex: images), or information that is provided to the model for training (ex: user logged-in time). If it is possible to provide the training data, it should also be provided. Training data is said to be available if the dataset used to train the model is linked or steps to reproduce it are provided in the model card.	Training Data. May not be possible to provide in practice. When possible, this section should mirror Evaluation Data. If such detail is not possible, minimal allowable information should be provided here, such as details of the distribution over various factors in the training datasets.
Ethical Considerations	20. Are ethical considerations discussed? 21. Is the process for considering ethical issues with the model card discussed? 22. Are concrete measurements to support the ethical considerations provided?	The ethical considerations section should address any form of bias or issues relating to the model's impact. Any steps that were taken to reduce these issues should be described. Ethical issues are often discussed in other sections of model cards. This is not a problem, as it is sometimes more appropriate to address them in other sections. For example, if there is potential bias in the training data, then discussing it in the section with the training data makes sense. More than anything, the ethical considerations section should show that the model creator did not ignore obvious ethical problems and took the chance to mitigate these where they could. If a model creator was not able to keep a model from having a bias towards one race, for example, we would rather them discuss that openly in this section than omit this section and hide it.	This section is intended to demonstrate the ethical considerations that went into model development, surfacing ethical challenges and solutions to stakeholders. Ethical analysis does not always lead to precise solutions, but the process of ethical contemplation is worthwhile to inform on responsible practices and next steps in future work.
Clarity	23. Clarity comments	Clarity refers to the overall organization and presentation of information in the model card. If a model card is not easily-readable, it is likely that users will skip over some very important sections to just grab some sample code. This defeats the whole purpose of the model card. Some model cards are left very sparse and a paper is linked that claims to have more information. This is not a model card that has good clarity because most users will not take the time to read an academic paper. Delegation of information to external sources should be kept at a minimum. There is a limit to how deep a model user will go before they give up on a model. If all of the information is kept under clear headers and is easily accessed, everyone will benefit.	